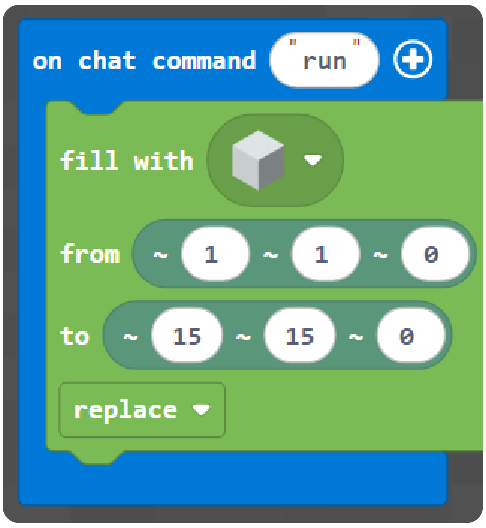
M003 Extension — English Worksheet (Advanced)

Topic: Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Week 3) · **Time:** about 45 minutes

A bonus picture challenge: copy **two bigger mobs** — a **pink axolotl** and a **black ender dragon head**. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 1, 0),
        pos(15, 15, 0),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

Big idea: (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

1 · Predict

Read each set of steps. Write what you will see **and the reason**.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

Three pink blocks share y = 13. Do they make a line going across or up? Which number stays the same, and why?

```
place black block at (8, 7)
place black block at (8, 8)
place black block at (8, 9)
place black block at (8, 10)
```

These four share x = 8. Are they a row going across, or a column going up? Which number changes, and why?

```
place pink block at (5, 6)
place pink block at (6, 6)
place pink block at (5, 7)
place pink block at (6, 7)
```

These four blocks change both numbers. What shape do they make on the wall, and why?

```
place pink block at (6, 10)
place black block at (6, 10)
```

Two blocks aim for the same (x, y). What happens on the wall — pink, black, or both? Explain the reason.

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — This should make the **axolotl's two eyes**, one at (6, 10) and one at (11, 10). Both land in the **same spot**.

```
place black block at (6, 10)
place black block at (6, 10)
```

Hint: the two eyes are at **different** x values — one is further across than the other.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The dragon's **eye** should sit at (8, 10). The two numbers are **swapped**, so it lands in the wrong place.

```
place purple block at (10, 8)
```

Hint: the **first** number is across (x), the **second** is up (y). Which one should be 8, which should be 10?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A friend stood on the **wrong block** and ran correct code. The axolotl came out **shifted two blocks too high** — every block landed at a y two bigger than the grid says.

```
(the code matched the grid exactly)
```

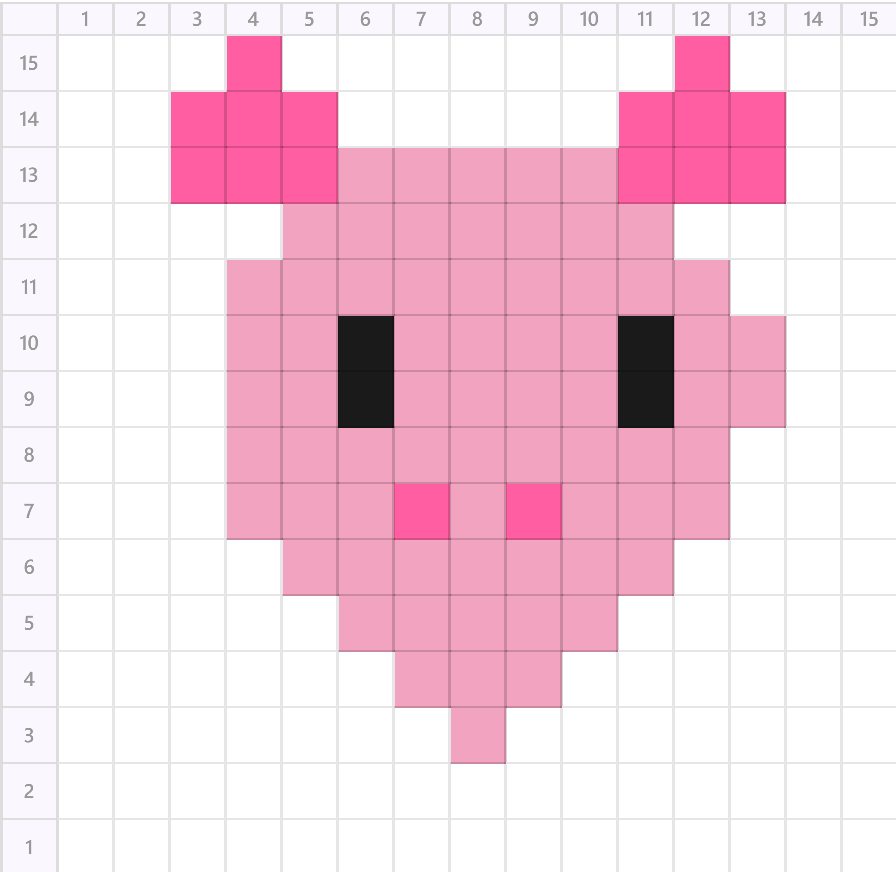
What did your friend forget to do before pressing run? Why does standing on the home spot change where every block lands?

3 · Show Your Work 📸 🎥

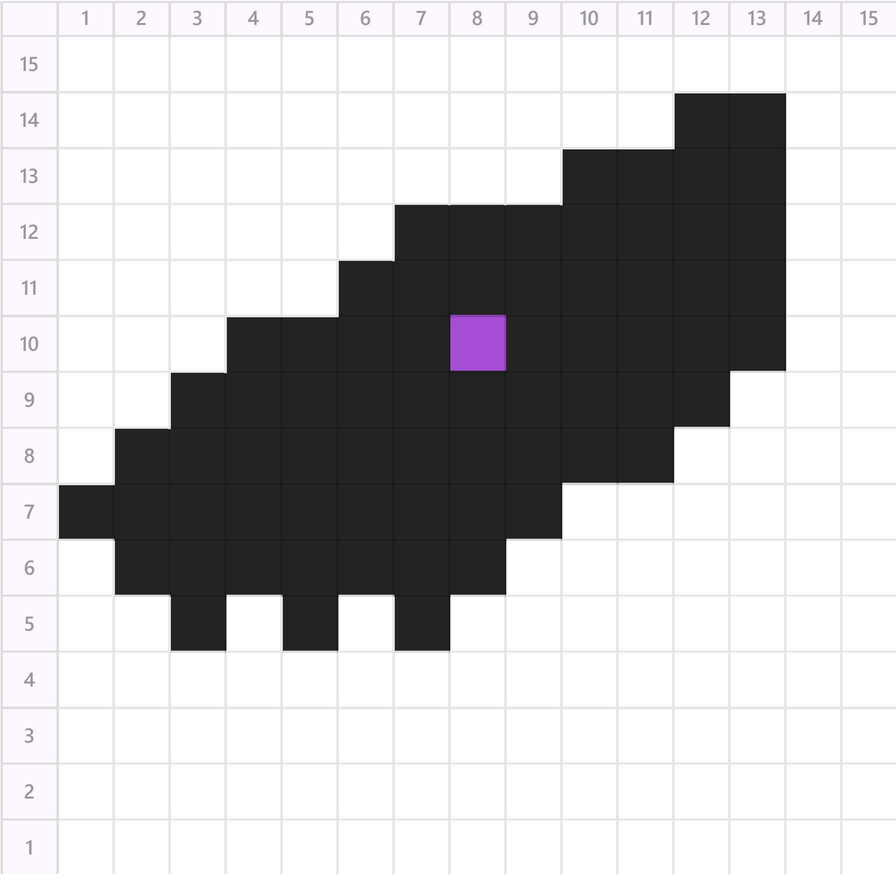
Stand on your **home spot** (the red block). Pick **one** mob and copy it block by block, reading each colored square's (x, y) — **x** across the top, **y** down the side.

Plan first. Before you place anything, find the **lowest** block and the **left-most** block of your mob and write their (x, y) here. That tells you where the picture starts.

🦜 Axolotl (pink · #f2a3c0, frills · #ff5fa2, eyes · black)



🐉 Ender Dragon (black · #232323, eye · purple #a64dd6)



Modify challenge. After you copy a mob exactly, change **one thing** that is your own idea — swap a colour, add a part, or make a row longer. Write the (x, y) of the blocks you changed, then build it.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.